

Fungal and Oomycete Effector Triage as a Benchmark Domain

Application of the domain-general framework · Five-agent construction plan · 15 September 2026

Protein science supplied the first application of the framework. Plant pathogen effector biology supplies a second, and it improves on the first in three respects that matter. The domain sits inside the BER mission rather than beside it. Brookhaven owns a foundation model in it. And the strongest published pipeline in the field states in its own discussion that it leaves multi-channel integration to a human. That last sentence is the decision this benchmark measures.

3.0 STANDING POSITION BEFORE ANY COMPUTE

Five findings carry the plan. Four of them close or reshape the design at zero cost, which is the point of the funnel.

1. **The natural item fails supply outright.** One secretome equals one item gives roughly 15 to 20 phylogenetically nested species against an $N_{\min}$ near 50. That cohort resolves an effect of about 0.076 in precision at twenty, so it detects only a swing four times larger than the one worth detecting. Panel items repair it. Species items become the external validity check, never the primary.
2. **The clustering measure is wrong for this domain.** Effector families are defined by fold, not by sequence. MAX, RALPH, ToxA-like, FOLD and LARS members share structure below detectable identity. Counting independent units at 40 percent sequence identity therefore overstates supply by an unmeasured factor. One Foldseek pass over predicted models settles it in a day, and nobody has run it.
3. **The mechanical floor already carries the effect most papers attribute to an agent.** PEACE reaches precision 0.217 against a chance rate of 0.011 with no agent anywhere in the loop, a twentyfold gain. Any agentic arm answers to 0.217, not to 0.011 and not to EffectorP's 0.041.
4. **Headroom is testable before a single model call.** PEACE argues that taxon-specific models beat an omnibus model. If that holds, per-lifestyle oracle channel weights beat global weights mechanically, and the size of that gap is the honest ceiling for an adaptive agent. If the gap falls below the smallest effect worth detecting, the task closes in week two with zero agent calls spent.
5. **A live cohort is available here and the field has already built one by accident.** Predector evaluated 35 effectors published after its own model froze. New validated effectors appear continuously, so a rolling post-cutoff stratum turns the permanent contamination disclosure of the protein case into a solved condition.

3.1 DOMAIN ENTRY

The entry rule says to map a domain's existing foundation model pipelines and then locate the points where a practitioner chooses under a constraint. Effector biology satisfies the first half abundantly, and the mission alignment falls out for free.

Mission fit

BER funds plant and microbe interaction science because pathogens set the yield ceiling on bioenergy feedstocks and because effectors reshape host-associated microbiomes and therefore soil carbon. Predector's own held-out evaluation set already contains *Melampsora larici-populina*, the poplar leaf rust. A program manager who asks why the lab studies insects does not ask why the lab studies poplar rust. PEACE carries DOE BER award KP1601011, which makes the instrument, the data and the funding line the same line.

Foundation models in the domain, by functional category

Category	Foundation model tools	Classical counterpart for arm 3
Sequence representation	ProtT5 and ProtTrans, ESM-2, PEACE, POOE, Fungtion	EffectorP 3.0, WideEffHunter, pepstats composition, HMMER against Pfam
Structure	AlphaFold 2 and 3, ESMFold, Boltz	Foldseek and TM-align against a curated effector fold library, HHpred
Secretion and localisation	DeepLoc, DeepSig, SignalP 6	SignalP 3 HMM, TargetP 2, Phobius, TMHMM
Genome context	Evo 2, Nucleotide Transformer, genomic LMs	RepeatMasker, OcculterCut AT-rich blocks, accessory contig calls, dN/dS
Fixed composites	PEACE ensemble, in house	Predector, fifteen tools plus a learning-to-rank model

Classical counterparts exist for every category, which is what condition 5 requires. Predector matters twice over. It serves as the strongest existing adjudicator, and it demonstrates that a fixed mechanical composition of this domain's tools already runs end to end at 4,514 proteins per hour on one sixteen-core node.

3.2 THE DECISION, AND THE SIX CANDIDATES

Candidate decision	The constraint	Who makes it
1. Screening queue. Which k secreted proteins from a newly sequenced pathogen enter the transient expression and hypersensitive response screen	Cloning and agroinfiltration capacity, roughly 20 constructs per batch	Every plant pathology lab that sequences a genome
2. Structural queue. Which k effectors go to beam time or cryo-EM	NSLS-II allocation	Structural biology groups, including Brookhaven's
3. Host target assignment. Which k host proteins to test against one effector	Pull-down and yeast two-hybrid throughput	Molecular plant-microbe labs
4. Taxon routing. Which taxon-specific predictor to apply	None that binds	Nobody, in practice
5. Secretion triage. Which proteins survive an ambiguous signal peptide call	Downstream pipeline cost	Pipeline authors
6. Surveillance. Which isolates to sequence in a feedstock disease watch	Sequencing budget	Surveillance programs

3.3 TIER 1, FREE FILTERS

Counting closes four of the six at zero compute, and it reshapes the fifth.

Candidate	Apparent scale	Scarce unit	Verdict
4. Taxon routing	thousands of secretomes	one classifier call decides the whole question	Fails 4. Lookup, not composition
6. Surveillance	thousands of isolates	no per-item label exists at all	Fails 3
3. Host target assignment	PHI-base at roughly 4,000 entries	validated effector and host target pairs, concentrated in about ten model systems	Fails 1. Amplification read as diversity
2. Structural queue	the deposited structural record	solved effector structures cluster into low tens of folds, and AlphaFold both proposes and scores	Fails 1 and 3 together
5. Secretion triage	10,000 to 20,000 proteins per genome	SwissProt-verified secreted fungal proteins, 256 train and 64 test in Predictor's partition	Fails 4 mostly. A prediction wearing a decision's clothes
1. Screening queue	9 pathogen proteomes held out by Predictor, plus MycoCosm	independent effector families after structural clustering, unmeasured	Survives, with the item redefined

Axis enumeration for the surviving task

Axis	Count available today	What it does to the design
Positive supply, sequence clusters	153 at 30 pct (Predictor), 212 (PEACE Fungtion), 685 at 40 pct (PEACE Fungi and Oomycete). Union unmeasured	Looks sufficient. Read alone, it misleads
Positive supply, structural clusters	not measured by anyone	The binding count. Fold families collapse sequence clusters by an unknown factor. One Foldseek run settles it
Negative supply	unlabelled, not verified	Predictor states plainly that its false positives may be real effectors. Precision becomes a lower bound with species-dependent bias
Contamination exposure	every positive predates every subject model	Static cohort fails condition 3 permanently. Live refresh is the only repair
Tool coverage	in planta RNA-seq exists for a subset of species only	The expression channel covers part of the cohort. C differs per item unless the design forces coverage

INSTRUMENT NOTE, RAISED CONSTRUCTIVELY

PEACE derives its Fungtion partition from Li and colleagues, who reduced sequence identity at 40 percent with CD-HIT and then filtered at structure level with ECOD. The newly curated Fungi and Oomycete partition applies CD-HIT at 40 percent and a strict BLASTP negative filter, which is the right choice for training. For benchmark purposes the unit count needs the domain's structural measure, because effectors diverge below sequence detection while holding fold. The 138 test positives are therefore 138 sequence clusters, and the number of independent units sits at or below that. Measuring the gap costs one afternoon and it protects the dataset rather than undermining it.

The item redefinition

One secretome as one item gives 15 to 20 correlated species. The panel construction restores supply without inventing data. Each item becomes a bounded candidate pool of n_{cand} equal to 500 proteins drawn from a single real pathogen secretome, seeded with 5 validated effectors whose structural clusters appear in no other panel. Roughly 120 panels follow from the available positive supply, which clears both N_{min} and the hundred-example bar on the project checklist.

Two design rules make the panel honest rather than convenient. First, decoys come from the same organism as the planted positives, so no arm can shortcut on lifestyle or taxon identity. Second, the unknown-positive rate in the background gets measured separately on saprotroph-decoy control panels, using the contrast Predictor already reports between pathogen and saprotroph score distributions. The absolute precision level stays a lower bound and

gets labelled one. The paired arm difference stays unbiased, because the deflation is common to every arm on the same panel.

3.4 TIER 2, THE MECHANICAL FLOOR, AND THE TEST THAT DECIDES EVERYTHING

Three floors, all built with no agent in the loop, all blinded. **Floor C, classical:** Predector itself, which is the strongest existing adjudicator. Running EffectorP alone as the comparator would price the gain against a straw man and violate the reporting rule. **Floor M, foundation model:** PEACE, ProtT5 embeddings with prototype-aware contrastive training, served from the Brookhaven checkpoint. **Floor M+, six channels:** Floor M plus fold-family match, genome context, in planta induction, and negative functional evidence such as CAZyme membership.

What the published record already fixes

Arm	Precision	AUPRC	Chance	Set
Chance	0.011	n/a	34 of 3,097	Function test
WideEffHunter	0.010	n/a	n/a	Function test
EffectorP 3.0	0.041	0.142	n/a	Function test
PEACE, two stage	0.217	0.558	n/a	Function test
EffectorP 3.0	0.037	0.067	138 of 7,038	Fungi and Oomycete test
PEACE, two stage	0.112	0.254	n/a	Fungi and Oomycete test
Predector, top 50	0.080	NDCG at 500: 0.928	n/a	Predector test, different partition

Two things follow immediately. The foundation model contributes a large, reproducible gain over classical methods with no agent involved, which is the effect that most agentic framings claim for their agent. And nobody has yet compared PEACE against Predector on a common partition, which makes that comparison both the floor the benchmark needs and a publishable result in its own right. It costs one afternoon of CPU time.

The headroom test

H1, THE HYPOTHESIS THE BENCHMARK TESTS

The optimal channel weighting depends on the item. Transposon proximity carries signal in a plastic necrotroph genome and none in a compact biotroph. RALPH homology dominates in *Blumeria*. Motif evidence dominates in oomycetes. A fixed mechanical ensemble must commit to one weighting. An agent can reweight per item. If that adaptivity is worth anything, it is worth the difference between oracle weights and global weights.

$$H = \text{precision_at_k}(\text{oracle weights}) - \text{precision_at_k}(\text{global weights})$$

KILL CONDITION, WEEK TWO

If H falls below delta, no agent can capture anything the mechanical composition leaves behind, and the task closes before any model call. If H exceeds delta, H becomes the honest ceiling for the agentic arm, and the benchmark reports the agent against that ceiling rather than against chance.

$$N_{\text{min}} = \text{ceil} \left((z_{(1-\alpha/2)} + z_{(1-\beta)})^2 \cdot \sigma_d^2 / \delta^2 \right) \quad \text{MDE}(N) = (z_{(1-\alpha/2)} + z_{(1-\beta)}) \cdot \sigma_d / \sqrt{N}$$

Taking a plausible advance estimate of sigma_d near 0.115 for precision at twenty, the numbers resolve as follows. A 120-panel cohort detects about 0.029, which sits below the smallest effect worth detecting. A species-level cohort of 18 detects about 0.076, which does not. Both numbers get published alongside every reported effect, and the second is the reason the species cohort serves as a generalisation check and never as the primary.

3.5 HYPERPARAMETERS, WITH DOMAIN VALUES

Symbol	Value	How the domain fixes it
k	20	One agroinfiltration batch. Cloning sets the ceiling, not the assay
n_cand	500	Panel size. Fixes chance at 0.04 and metric granularity at 0.05
delta	0.05	One additional validated effector per batch. Each one anchors a resistance-gene lead, so the action it governs carries high downstream value
sigma_d	measure it	Paired per-item difference between Floor C and Floor M+. Zero model calls
N_min	about 45 to 55	Derived, never asserted. Recompute once sigma_d lands
tau	TM-score 0.50	Structural, with CD-HIT 40 percent as the secondary axis. Report sensitivity at 30 and 50 percent identity
C	measure, expect 4	Six nominal channels. Verify mechanically at C minus 1 and C. Sequence composition and pLM likelihood are near neighbours, so the honest count runs lower than six
S	start at 5	Set by the coverage curve. Report coverage at S as a curve, never coverage at 1
rho	ceiling minus floor > delta	Condition 2 in numeric form, evaluated as H above

3.6 TIER 3, WHAT THE INSTRUMENT WILL DO TO US

Class	The trap, stated concretely	The rule that prevents it
II. Instrument	pLDDT as an effector gate. Effectors are small, cysteine-rich and disulfide-stabilised, and structure predictors return low confidence on exactly that profile. A structural channel that filters on pLDDT discards the positives it exists to find	Score fold-family match with Foldseek against a curated effector fold library. Treat pLDDT as a covariate, never as a gate
II. Instrument	Licence substitution. SignalP, TMHMM and Phobius carry academic licences. Substitutes change the measured quantity. Predector reports signal peptide recall on known effectors ranging from 84 percent for DeepSig to 92 percent for TargetP 2, an eight point spread that exceeds delta	Justify every substitute against the quantity the task scores on, never against the tool it replaces. Clear licences in week three, not week twelve
II. Instrument	Served checkpoint drift. ProtT5-XL against XXL, ESM-2 at 8M against 650M, EffectorP 3.0 against 2.0, AlphaFold 2 against 3 against Boltz, Foldseek mode and version	Hash the served weights, record parameter counts, verify from the model's side rather than from configuration
III. Gate and I. Supply	Label circularity through lineage. PEACE and Predector positives both descend from EffectorP training curation, and that curation used the same small-secreted-cysteine-rich heuristics the tools encode	Restrict labels to functional evidence, meaning hypersensitive response assay, knockout virulence phenotype or validated host-target interaction. Computational evidence codes in PHI-base do not qualify

3.7 TIER 4, HARNESS CONTROL FIRST, NOT FOURTEENTH

The single most expensive omission in the protein record was running the known-good harness control last. Here it runs on day one of tier 4, on one panel, with a capable model through the full tool surface. It converts every later failure from ambiguous to localised.

One prompt-level trap transfers directly. Supplying genome and proteome file paths is necessary for a tool-using arm and is fabrication bait for a tool-free one. Expect arm 1 to invent PHI-base accessions, cysteine percentages and SignalP scores rather than declining. Rule on prompt symmetry against prompt neutrality explicitly and in advance, and record the ruling in the amendment ledger.

3.8 ARMS, CONTRACT AND METRICS

Arm	Grant	What it bounds
1. LLM alone	panel FASTA and organism name, no tools	Contamination. A frontier model performing at chance on public sequences bounds the concern better than any audit
2. Mechanical floors	Floor C, Floor M, Floor M+, zero calls	The bar. Also supplies sigma_d and H
3. LLM plus classical	every classical tool, foundation models withheld at the firewall	Isolates the foundation model contribution. This is the arm the field never runs
4. LLM plus classical plus FM	everything	The claim under test

The contract asks for exactly twenty accessions, ranked, each carrying a one-line evidence statement that names the channels consulted, plus at least one explicit rejection with its reason. Precision at twenty is the primary metric and the ordering is fixed before the run, because in the protein case the decision metric and the ranking metric disagreed in sign on the same data. Coverage at S accompanies it as a curve. Process metrics, meaning channels consulted per candidate, explicit rejections and handling of channel disagreement, get their own denominators and remain measurable on arms that never submit a valid answer. The scaffold stays fixed across arms and gets disclosed, since a plan-act-observe loop and a coding-agent scaffold are different subjects on the same task.

3.9 THE FIVE-AGENT CONSTRUCTION PANEL

Panel roles pay only when they hold different information. A critic sharing the proposer's context catches reasoning errors. An agent holding the filesystem, the traces and the served configuration catches factual errors, and in construction work the factual errors are both more numerous and more costly.

Agent	Holds	Owns	Catches
1. Curator	PHI-base, UniProt, MycoCosm genomes, RNA-seq archives, CD-HIT, Foldseek	Tier 1 axis enumeration in one pass. Structural clustering. Panel construction with disjoint clusters	Supply failures. Refuses to report any single count alone
2. Instrument	the filesystem, the served endpoints, the tool source code	Tool cards written from code first and checked against the paper second, with disagreements emitted rather than reconciled. Execution-time provenance hashing	Wrong-quantity tools, undersized checkpoints, parameters that do not exist, manifests that diverge from what the runner sends
3. Floor	the fixed tools and the held-out design split only	Blinded Floor C, M and M+. sigma_d, N_min, MDE. The oracle-weight headroom test. Item-design search on the design split	Tasks with no room. Runs before any agentic arm exists
4. Referee	the preregistration, the amendment ledger, the scorer. Never sees arm identity	Scoring, metric primacy, coverage semantics, repair budget held identical across arms	Silent metric reversal, unequal repair budgets, unscored answers sitting on disk
5. Adversary	the traces, the gate declarations, the literature	Must-fire and must-not-fire controls generated per gate. Threat-stated stop conditions. Harness control. Clause-by-clause constitution compliance	Gates firing on proxies, name-based stop conditions, surface failures misread as subject failures

Agents 1 and 3 between them decide whether the benchmark exists, and neither spends a model call to do it.

3.10 THE FUNNEL, WITH THIS DOMAIN'S TRAFFIC

Tier 1, free. Conditions 1, 3, 4. No compute. Week 1 Six candidates enter. Taxon routing dies as lookup. Surveillance dies for want of labels. Host targets and the structural queue die on supply. Secretion triage survives as a prediction and gets set aside. One task continues, with its item redefined.
Tier 2, cheap. Conditions 2, 6. One afternoon of CPU. Week 2 Three blinded floors. PEACE against Predector on a common partition. sigma_d, N_min and MDE published. The headroom test rules. This is the real gate.
Tier 3, weeks. Condition 5. Weeks 3 to 5 Tool cards from code. Licences cleared. Checkpoints hashed. Containers built on Predector's existing Nextflow and conda packaging. Construct validity checked before functional certification, not after.
Tier 4, harness. Known-good model first. Week 5 One panel, one capable model, full surface. Any failure here indicts the surface, never the subject.
Run. Four arms. Weeks 6 to 8 120 panels, preregistration frozen, referee blinded to arm identity.

3.11 AGAINST THE PROJECT BENCHMARK CHECKLIST

Column	Status	Note
Domain we care about	Pass	BER plant and microbe interaction, feedstock disease, in-house model, in-house award
100 or more examples	Pass	About 120 panels. Species-level items reach only 18 and fail this column, which is why they are not primary
Relevant foundation model exists	Pass	PEACE and ProtT5 in house, plus ESM-2, AlphaFold 3 and Evo 2
Containerised in Harbor	Buildable	Predector already ships Nextflow, conda and Docker. Licence-restricted tools need substitution or a site licence, decided in week 3
Frontier accuracy between 10 and 70 percent	Measure at tier 4	Chance sits at 0.04 and the floor near 0.25 on this panel geometry. The window is plausible, and nobody should promise it before the harness control runs
Failures are genuine, not verifier artefacts	Owned	Agent 5, with must-fire gate controls and the harness control. The protein case lost an entire arm to a wiring failure, and this column exists to prevent that

3.12 KILL CONDITIONS, WRITTEN BEFORE THE RUN

1. Headroom H falls below delta at tier 2. Close the task. Zero agent calls spent.
2. Structural clustering leaves fewer than 45 independent panels. Close, or restate delta at 0.10 and say so in the abstract.
3. Mechanical channel ablation leaves C at two or fewer. Close. The last channel adding nothing without an agent adds nothing with one.
4. Labels cannot be sourced independently of the EffectorP curation lineage. Close, because condition 3 is not negotiable.
5. The live post-cutoff stratum yields fewer than 20 clean items per refresh. Keep the benchmark, drop the clean-contamination claim, and disclose.

3.13 WHAT THIS DOMAIN ADDS THAT PROTEIN SCIENCE DID NOT

The composition step is documented as a human step. Predector's discussion recommends that users supplement its rankings with selection profiles from pan-genome comparison, presence and absence variation, association studies, differential expression and genomic landscape features such as distance to repeats and telomeres. The tool's own authors describe that integration as manual. A benchmark that measures exactly that integration is not a constructed task. It is the task the field already performs by hand, which is the strongest possible answer to condition 4.

The contamination problem has a live solution the field has already prototyped. Predector's post-development evaluation on 35 newly published effectors is a live cohort in everything but name. Formalising the refresh converts a permanent disclosure into a satisfied condition.

The floor is strong and honest. In this domain the foundation model floor beats the classical floor by a wide, reproducible margin with no agent anywhere. That makes the benchmark hard in the right way. Any agentic gain reported here will have cleared a bar that most published agentic claims never face.

Companion documents: the domain-general framework, Section 1, and its protein science application, Section 2. Source material for this section: PEACE preprint (Dai, Lin, Yoo and Liu, Brookhaven, April 2026, DOE BER KP1601011) and Predector (Jones and colleagues, Scientific Reports 2021). All performance figures quoted above come from those two papers and carry their partitions with them. No figure here has been recomputed on a common partition, which is itself the first deliverable.